

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

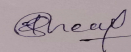
ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	Total Marks:	20 marks
CO2 Statement:	Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.		
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Section / Batch:		Date:	27.07.2026

Code of Conduct

I Rhema Daphne E G (192321097) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis (BL4)	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	100 Marks

← Mobile IP Analysis

Packet Tunneling & Encapsulation

Visualizing the secure transport of IP-in-IP packets from a Correspondent Node (CN) to a Mobile Node (MN) via the Home Agent (HA) and Foreign Agent (FA).

Origin

CN

Src: 192.0.2.100
Dst: 203.0.113.5

Encapsulation

Home Agent

HA

Decapsulation

Foreign Agent

FA

Destination

MN

Id: 192.0.2.10

← Mobile IP Analysis

Architectural Overview

Mobile IP enables seamless network roaming. This diagram illustrates the relationship between the Home Network, Foreign Network, and the routing mechanics required when a Mobile Node (MN) relocates while maintaining ongoing sessions with a Correspondent Node (CN).

HOME NETWORK

Home Agent

HA

Internet

FOREIGN NETWORK

Correspondent

CN

Foreign Agent

FA

← Mobile IP Analysis

Registration Process

Step-by-step timeline of the Mobile Node (MN) registration with the Home Agent (HA) via a Foreign Agent (FA).

1. Agent Advertisement

Success

The Foreign Agent (FA) periodically broadcasts its presence on the local network. The Mobile Node (MN) receives this advertisement and determines it has a foreign network.

ICMP Router Discovery Protocol (RSDP)

2. Registration Request

Success

The MN sends a Registration Request to the FA, which forwards it to the Home Agent (HA). This request includes the MN's home address and the FA's care-of address (COA).

Source: 192.168.2.10 (MN COA)
Dest: 10.0.1 (HA)

3. Registration Reply

Pending

The HA processes the request, creates a binding, and sends a Registration Reply back to the FA, which then forwards it to the MN, completing the process.

← Mobile IP Analysis

Triangular Routing Latency

ASes

ISPs

Sinks

Direct

Triangle

Optimized

Optimal/Optimized

Sub-optimal

Packet Loss (Handoff)

Pre-Registration

2.4%

During Handoff

5.8%

Post-Registration

0.1%

Active Optimizations

Route Optimization

Enabled (RFC 6275)

Active

← Mobile IP Analysis

Protocol Explorer

Mobile IP allows location-independent routing of IP datagrams on the Internet. Each mobile node is identified by its home address, regardless of its current point of attachment to the Internet.

Start Module >

HA Status ONLINE

FA Status ONLINE

MN Node ROUTING

Modules

Architecture

Components and addressing...

Registration

Request and reply messaging flow.

Tunneling & Encapsulation

Deep dive into IP-in-IP, tunnel encapsulation, and GRE protocols.

Analysis

Performance and packet tracing.

Latency Docs

Optimization strategies.

Live Topology Preview

Expand

Overview

Process

Tunneling

Analysis

Architecture Comparison

Analyze fundamental differences between centralized and decentralized topologies.

Cellular Network

Infrastructure

A highly organized, centralized topology relying on fixed physical infrastructure to manage communication between mobile nodes.

Infrastructure Dependency

High. Requires permanent base stations (eNodeB/gNodeB), core network servers, and physical backbone (fiber/microwave) to function. Failure of a base station creates a complete coverage hole.

Centralized Control

High Capacity

VS

MANET

Decentralized

Mobile Ad Hoc Network. A self-configuring, infrastructure-less network of mobile devices connected dynamically.

Infrastructure Dependency

Zero. Nodes act as both end-systems and routers. The network forms dynamically without relying on any pre-existing infrastructure, providing high resilience in austere environments.

Peer-to-Peer

Self-Healing

Dynamic Topology

Performance Metrics

Comparing Cellular vs MANET architectures.



Architecture Insights

Cellular maintains high stability and centralized control, while MANET offers superior resilience at the cost of broader scalability.

Scalability

Network growth capacity

Cellular



MANET



Fault Tolerance

Resilience to node failure

Cellular



MANET



Mobility Support

Dynamic topology handling

Cellular



MANET



Home

Routing

Analysis

Scenarios

Application Scenarios

Strategic deployment models for Cellular and MANET architectures, highlighting optimal use cases and trade-offs.

Cellular Infrastructure



Public Telecommunication

High Density

Global voice and data networks providing high-bandwidth, reliable connectivity to mass consumer markets via fixed infrastructure.



Smart Cities (IoT)

Low Latency

Centralized control arrays managing autonomous traffic systems, environmental sensors, and municipal utilities.

Mobile Ad-Hoc Networks



Home

Routing

Analysis

Scenarios

Routing Mechanisms

Comparative analysis of infrastructure routing topologies: Cellular vs. Mobile Ad-hoc Networks (MANET).

Cellular Infrastructure

Centralized / Hierarchical TOPOLOGY

Static



MANET Infrastructure

Distributed TOPOLOGY

Dynamic



Metric

Cellular vs MANET

Infrastructure

Fixed vs. Ad-hoc

Overhead

Low vs. High

Scalability

High vs. Limited

Protocol Viz

Proactive Protocol Analysis

OLSR

Optimized Link State Routing (OLSR) maintains a proactive route table, ensuring routes are instantly available when needed. This approach minimizes setup delay but increases continuous control traffic overhead.

Route Discovery & Maintenance

HELLO Message

T=5ms

Broadcast

Nodes periodically broadcast HELLO messages to discover 1-hop and 2-hop neighbors. This continuous chatter establishes local neighborhood connectivity.

MPR Selection

T=50ms

Each node selects a subset of its neighbors as Multipoint Relays (MPRs). Only MPRs are responsible for forwarding broadcast messages, significantly reducing control traffic floods.

Topology Control (TC)

T=200ms

Broadcast

MPRs generate and broadcast TC messages containing link-state information to the entire network. This builds the global topology map.

Routing Table Calculation

Continuous

Nodes use Dijkstra's algorithm on the topology map to compute shortest paths to all destinations, keeping the routing table proactively updated.

Performance Insight

Route Setup Delay

~6 ms

Routing Overhead

High

Trade-off: Minus: more latency for data

Protocol Viz

Network Routing Architecture

Explore the fundamental paradigms of ad-hoc network routing. Understand how proactive and reactive protocols manage topology data and route discovery in dynamic environments.

Start Analysis

Proactive (OLSR)

Table-Driven

Optimized Link State Routing maintains fresh lists of destinations and their routes by periodically distributing routing tables throughout the network. Ideal for static topologies requiring immediate route availability.

Constant Overhead Zero Latency

Reactive (AODV)

On-Demand

Ad hoc On-Demand Distance Vector routing finds a route only when desired by the source node. It reduces baseline network overhead but incurs a delay during the route discovery process. Ideal for highly dynamic networks.

Low Overhead Discovery Latency

Overview Proactive Reactive Compare

Protocol Viz

Reactive Protocol Analysis

LIVE AODV

Ad hoc On-Demand Distance Vector (AODV) establishes routes only when necessary, minimizing broadcast overhead but increasing initial latency.

Route Discovery (RREQ / RREP)



Initial Delay Alert

Significant latency observed during connection setup. Route discovery must complete before data transmission begins.

Setup Latency: +450s

Performance Insight

Control Overhead Low (12%)

Minimal background traffic when routes are inactive.

Initial Delay High (~450s)

Route discovery required before first packet transfer.

Run Simulation

LIVE EVENT STREAM

[10:42:01] Node 4 Initiated RREQ
[10:42:02] Node 7 Forwarded RREQ
[10:42:03] Node 12 Forwarded RREQ

Overview Proactive Reactive Compare

Protocol Viz

Comparative Summary

An analytical breakdown of Proactive vs. Reactive routing protocols across key performance metrics.

Metric Analysis

Metric	Proactive	Reactive
Routing Overhead	↑ High	↓ Low
Initial Latency	↓ Low	↑ High
Scalability	Poor in large nets	Better in large nets
High Mobility Perf.	Struggles with static routes	Adapts quickly

Mobility Impact

Proactive Reactive

Efficiency score under frequent topology changes.

Expert Recommendation

The choice between proactive and reactive routing depends fundamentally on the operational environment's

Overview Proactive Reactive Compare

VitalCare Mobile IP

VitalCare Mobile IP

VitalCare Mobile IP

VitalCare Mobile IP

Healthcare Hub

Real-time status of connected healthcare nodes across the mobile IP network.

2

Active Ambulances

(+)

15

Remote Doctors

LIVE

124

Connected Monitors

Network Topology & Handoffs

FULL SCREEN



Handoff Analysis

Ambulance-04: Handing over to City-Central Base Station

Handoff Timeline



Latency Spike
185 ms



Packet Loss
2.4 %

Detected during transit phase. Retransmitting...

Security Monitor

Identifying and mitigating security risks in Mobile IP handoffs.

Protocol: IPsec Enabled
High Protection Active
Tunnel Status
Secure (MN to HA)

Encrypted Path vs Unencrypted



Risk Assessment



Active Threat Vectors

Traffic Interception
High Risk
Vulnerability detected in unencrypted Binding Updates. Malicious node attempting to intercept data destined for MN.

Review Logs

Binding Update Hijacking

Elevated
Attempted spoofing of MN's Care-of-Address (CoA) to redirect traffic. HA currently dropping unauthenticated BU packets.

Network Optimizer

Proposed design improvements for system reliability.

Active Configurations

Smooth Handoff Buffering
Prevents packet loss during AP transitions.

Route Optimization (RFC 6275)
Direct CN-to-MN routing bypassing HA.

Hierarchical Mobile IP
Localizes signaling for micro-mobility.

Performance Impact



Critical hardware update available for edge nodes.

Upgrade Foreign Agent Buffers +

Hub Handoff Security Optimize

Network Hub

Configure core scenario parameters to initiate network logic simulation.

Scenario Configuration

ID: SCEN-8472

NODE MOBILITY

Slider: Static to Highly Mobile

INFRASTRUCTURE AVAILABILITY

Slider: Zero (No Links) to Total (Full)

RELIABILITY REQUIREMENTS (SLA)

99.9% Standard, 99.99% High, 99.999% Mission Critical

Reset

▶ Simulate

Hub Analyze Decision Guide

Network Profiler

Architecture trade-off analysis & parameter sensitivity

📄 Export Report

💾 Save Profile

Architecture Trade-offs

Recommended Baseline



1

The recommended Hybrid Edge-Core Model sacrifices 15% raw throughput for a 40% gain in mobility and 25% reduction in

Hub Analyze Decision Guide

Decision Engine

output

Final topology recommendation computed from input heuristics, load balancing requirements, and scalability thresholds.

OPTIMAL TOPOLOGY

Cellular Infrastructure

Recommended for high mobility and broad coverage requirements.

CONFIDENCE SCORE 94 %

Justification Report

Routing Complexity

Hub Analyze Decision Guide

Deployment Guide

Practical implementation steps for Ad Hoc Mesh Network architecture.

Key Architecture

TOPOLOGY: MESH



Resource Requirements

@ Hardware Transceivers 802.11s eq, 34V 1K7 / 1W

Hub Analyze Decision Guide